

TrueAI: An AI-Enhanced Intervention to Prevent Fake News Spread Among older adults on Social Media

Leijing Zhou*
Zhejiang University
Hangzhou, China
leijing@zju.edu.cn

Xiaotong Guan
Zhejiang University
Hangzhou, China
22451362@zju.edu.cn

Peigen Liu
Zhejiang University
Hangzhou, China
peigenliu@zju.edu.cn

Abstract

The proliferation of fake news on social media presents a significant societal challenge, particularly for vulnerable populations such as older adults. This paper introduces TrueAI, an AI-enhanced intervention founded on Technocognition. TrueAI integrates cognitive principles, intervening at the critical moment that older adults attempt to forward potential fake news. By presenting real-time fake news alerts and comparisons with verified information from authoritative sources, TrueAI facilitates informed decision-making. Preliminary experiments (N=15) showed that TrueAI significantly reduced participants' forwarding of fake news. As a user-centric intervention leveraging AI, TrueAI offers the CSCW community valuable insights into designing socio-technical solutions to mitigate the negative impacts of fake news on vulnerable groups.

CCS Concepts

• **Human-centered computing** → **Social media**; **Human computer interaction (HCI)**.

Keywords

Fake News; Older Adults; AI-enhanced Intervention; Social Media; Digital Media Literacy

ACM Reference Format:

Leijing Zhou, Xiaotong Guan, and Peigen Liu. 2025. TrueAI: An AI-Enhanced Intervention to Prevent Fake News Spread Among older adults on Social Media. In *Companion of the Computer-Supported Cooperative Work and Social Computing (CSCW Companion '25)*, October 18–22, 2025, Bergen, Norway. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3715070.3749243>

1 Introduction

Social media has become a vital channel for information dissemination, but it has also increasingly become a breeding ground for fake news [30]. Fake news refers to intentionally fabricated or distorted information, presented in a news format, and designed to mislead the public, potentially causing negative social, political, or economic impacts [2]. With the rise of generative artificial intelligence, the scale, realism, and targeting of fake news have intensified, making its spread on social platforms more rapid and widespread [25].

*Corresponding author

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CSCW Companion '25, Bergen, Norway

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1480-1/25/10

<https://doi.org/10.1145/3715070.3749243>

Notably, older adults have emerged as one of the most vulnerable user groups to the spread of fake news [22]. The speed at which fake news spreads, six times faster than true news [28], significantly amplifies their susceptibility. Research indicates that older adults are more likely to visit dubious news websites [15]. Individuals aged 65 and older share fake news at seven times the rate of younger adults [14], and have been categorized as super-spreaders of fake news on Twitter [13]. However, this heightened susceptibility to fake news cannot be simply attributed to a lack of discernment, but rather to the complex interplay of cognitive decline, social changes, and digital illiteracy [8]. From a cognitive psychology perspective, older adults are more prone to biases such as reduced skepticism and overreliance on familiar sources, which may weaken their vigilance toward information credibility [23, 27]. From a social psychology standpoint, social isolation may motivate older adults to share unverified news as a means of establishing or maintaining social connections [26]. Furthermore, varying levels of digital literacy can hinder their ability to utilize tools designed to verify information credibility.

Existing automated fake news detection algorithms are advanced [1], enabling efficient processing of large-scale information flows, but they often intervene only after fake news has spread [24]. Furthermore, their one-size-fits-all nature neglects the unique cognitive and social motivations affecting older users [7, 8]. Beyond the limitations of these technological approaches, digital literacy training often involves slow learning processes, imposes high cognitive demands, and is detached from real-world contexts, making it difficult to apply effectively in real-time online scenarios. Therefore, considering the limitations in both current technical detection and traditional digital literacy training, older adults still lack effective, real-time intervention and support when facing fake news.

To address the aforementioned issues, we proposed and evaluated TrueAI, an intervention designed to provide proactive support while considering the cognitive characteristics and social motivations of older adults. TrueAI utilizes Technocognition, moving beyond traditional detection and generic training. At crucial moments when older users attempt to forward news, the system offers specific and responsive interventions. This includes warning them of potential risks, explaining the reasons it is considered fake news, providing a comparison between the fake news and authoritative news, and offering options to forward authoritative news instead. This not only aims to enhance users' ability to critically evaluate news but also actively responds to and guides their healthy social sharing behaviors. Preliminary evaluations indicated that TrueAI significantly reduces the forwarding of fake news by older adults and effectively promotes their digital media literacy. This research is dedicated to exploring a user-centered, proactive intervention

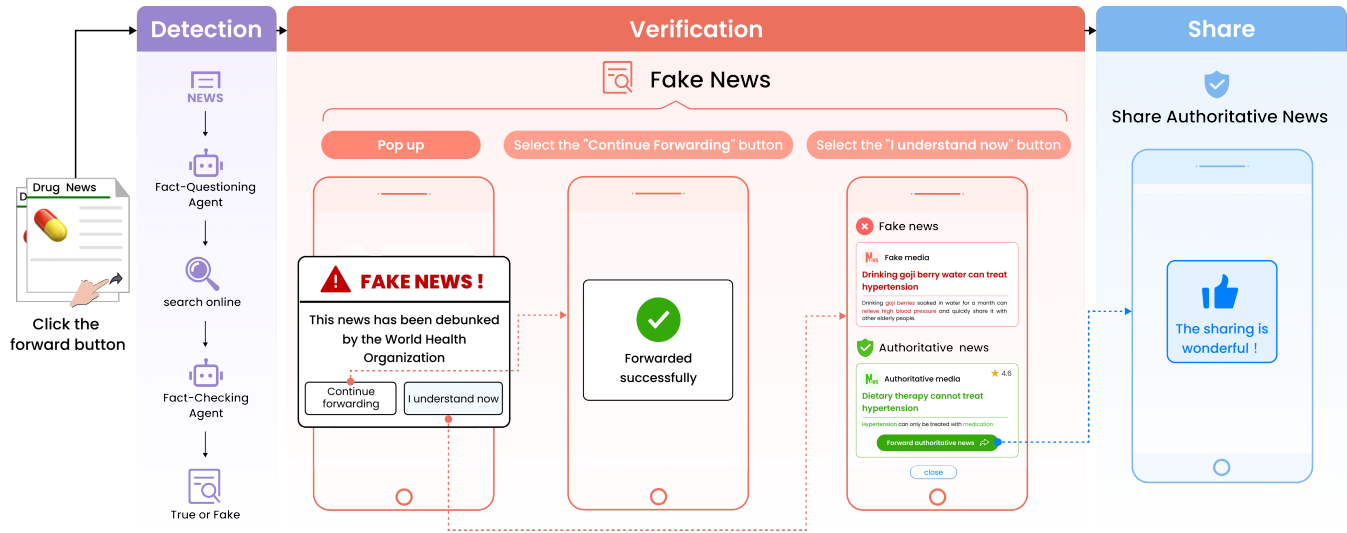


Figure 1: TrueAI architecture

paradigm, aiming to provide a more effective solution for tackling the problem of fake news faced by older adults.

2 Related Work

2.1 Fake News Detection Technology and Intervention Strategies

In response to the growing problems of fake news on social media, existing research has contributed various strategies. Machine learning-based algorithms [1] enable large-scale automated detection, crowdsourced fact-checking [3] utilizes collective intelligence for content verification, and fact-checking labels [16] directly alert users to information risks. These approaches have demonstrated significant effectiveness in identifying and curbing fake news. However, platform algorithms, driven by the pursuit of maximizing user engagement, may prioritize recommending and disseminating content that is more sensational rather than accurate, inadvertently fueling the spread of fake news [24]. Most of the existing technological measures intervene after news dissemination, and they cannot effectively prevent users from sharing this critical link in the dissemination process.

Interventions such as warning labels [21], lateral reading [29], reverse image search [9], and digital literacy curricula [6] have shown some effectiveness in helping users discern the authenticity of news. Research has shown that most existing warning labels are designed to be small, which can cause users to ignore such prompts or have difficulty accurately understanding their meaning [4, 5]. MediaWise [22] aims to improve digital literacy by teaching older adults techniques such as lateral reading and reverse image search. These methods are often slow to learn and cognitively demanding, reducing their appeal to older adults.

Due to the tendency of older adults to rely on heuristic judgments and lack systematic analytical skills [7, 8], the overall effectiveness of these interventions in this group is often limited, failing

to address the cognitive and social motivations behind older adults' behavior.

2.2 Technocognition

Technocognition, advocates for a "cognitively-inspired design of information architectures" [10, 18, 19]. It posits that synthesizing insights from cognitive science with targeted interventions in digital architectures enables the design of effective technological countermeasures against fake news [17]. Its core principle is to treat the information architecture itself as a primary intervention mechanism, designed to nudge users towards a more analytical and reflective mode of thought at critical moments. Fazio [12] found that when users were prompted to pause and explain why a headline might be true or false before sharing it, their intention to spread false news decreased significantly. This demonstrates that architecturally-embedded interventions can serve as potent, scalable countermeasures to fake news.

3 TrueAI

This section outlines the design of TrueAI, a system to reduce the spread of fake news by older adults on social media. Based on technocognition, TrueAI uses low-friction interventions to foster deliberate information processing and positive online interactions. The step has two core functions: (1) providing timely and cognitively-aware alerts at critical forwarding points to counter fake news, and (2) promoting the sharing of verified, authoritative news. The TrueAI process is mainly divided into three steps: first detection, then verification, and finally share, as shown in Fig. 1.

3.1 Detection

The detection first uses a Fact-Questioning Agent to generate questions based on the news content, then employs a Fact-Checking Agent to evaluate these questions by incorporating externally retrieved evidence, ultimately determining the news' authenticity.



Figure 2: Screenshot of TrueAI screen interface. (a) News headlines. (b) News body. (c) Fake news prompt. (d) Brief fact check summary. (e) "Continue forwarding" button. (f) "I understand now" button. (g) Fake news headlines. (h) Annotate the content of fake news. (i) Authoritative news headlines. (j) Annotate the content of authoritative news. (k) "share authoritative news" button. (l) Positive feedback.

3.2 Verification

When older adults attempt to forward content identified as potentially fake news, TrueAI's intervention mechanism triggers a subtle, low-friction pop-up. It initially displays a clear warning message (e.g., "This news could be fake news. Please reconsider.") and a concise fact-check summary from an authoritative source (e.g., "The World Health Organization has debunked this claim."). To minimize cognitive load and support informed decision-making, the information follows progressive disclosure principles. Initially, only the core reason for the warning is revealed, and users can then opt to access a more detailed comparative analysis and the full fact-check report. Furthermore, the pop-up's cognitively-aware design uses visually salient cues, like cautionary red tones, making the warning promptly noticed and easily understood, while minimizing undue alarm and disruption.

3.3 Share

TrueAI's "share authoritative news" feature constructively satisfies users' social needs with reliable information. When a user attempts to share potential fake news, the system not only offers certified authoritative news as an alternative but, more importantly, provides positive feedback for any user action of proactively sharing authoritative content. This technocognition-based reinforcement aims to guide more responsible and sustainable sharing habits.

4 Evaluation of TrueAI

We conducted a preliminary study to assess the effectiveness of TrueAI in reducing fake news forwarding. Due to limitations with accessing social media platform APIs and the need for controlled conditions, we developed a simulation environment replicating the

core functionality of WeChat, China's most popular social media, as shown in Fig. 2.

4.1 Participants

A total of 15 participants were recruited through local community centers. Participants included 7 women and 8 men, aged between 61 and 76 years ($M = 69.47$, $SD = 4.36$). All participants used at least one social media platform daily (e.g., Facebook, WeChat). The study was conducted using smartphones. Participants interacted with TrueAI, which featured 5 embedded true news items and 5 fake news items. Data collection involved questionnaires and interviews. Each session lasted 30–40 minutes, and each participant received \$5 as compensation.

4.2 Experiment Design

First, researchers provided participants with a detailed explanation of the study's purpose and procedures, and participants signed an informed consent form. They were explicitly informed that their data would be anonymized and that they could pause or withdraw from the experiment at any time. As shown in Fig. 3, Participants interacted with the TrueAI for 15–20 minutes, during which the system backend logged each instance of news-sharing behavior. Researchers did not intervene or observe during the interaction. Afterward, participants completed the System Usability Scale (SUS)[20] to evaluate the usability of TrueAI. Finally, a 5–10 minute semi-structured interview was conducted to gather user feedback, cognitive changes, and experiences with the TrueAI. All interviews were audio-recorded with verbal consent and used for subsequent transcription and analysis.



Figure 3: Participants conduct experiments using TrueAI

4.3 Data Analysis

Usability was assessed using a standardized SUS scoring procedure, and descriptive statistics were used to summarize participants' overall perceptions of TrueAI. Qualitative insights were gathered through semi-structured interviews, analyzed using thematic analysis [11]. Two researchers independently conducted open coding of the transcribed data, followed by iterative refinement to resolve overlaps and ensure relevance. The resulting themes captured participants' perspectives on the effectiveness, usability, and cognitive impact of TrueAI.

4.4 Quantitative Results

4.4.1 Forwarding probability. To evaluate the impact of TrueAI intervention reminder on older adults' fake news forwarding behavior, this study employs the Chi-square test to analyze the behavioral changes before and after the intervention. As shown in Table ??, the Chi-square test reveals a statistic of 43.081, a p -value of 5.253×10^{-11} , and 1 degree of freedom, with the p -value significantly lower than 0.05. Additionally, the Phi coefficient is 0.54, indicating a large effect size according to Cohen's standards. The 95% confidence interval (CI) for the Risk Difference is [-0.589, -0.344], showing a 34.4% - 58.9% reduction in forwarding behavior due to the intervention with 95% confidence. Results highlight the significant impact of TrueAI in altering older adults' fake news forwarding behavior.

Table 1: Chi-square Analysis of Behavioral Changes Before and After Intervention

Behavior	Non-intervention	Intervention	Total	χ^2	p -value
Not forward	40	75	115	43.081	5.253×10^{-11}
Forward	35	0	35	–	–
Total	75	75	150	–	–

4.4.2 SUS Score. SUS evaluation with 15 older adults revealed good usability for TrueAI ($M = 79.5$, $SD = 3.98$), indicating its interface and interaction design are well-aligned with their expectations.

4.5 Qualitative Results

System Effectiveness: All participants unanimously acknowledged the value of TrueAI's fake news warning feature and stated that it enhanced their confidence during the information dissemination process. Unlike traditional alert mechanisms, TrueAI not only identifies and blocks the forwarding of potential fake news but

also provides comparative analyses between true and fake news, along with authoritative source links for sharing. As participants (P5, P12) noted, "TrueAI's prompts are very helpful. They can stop me in time and explain the reason. It would be even better if it could recommend more authoritative news."

System Usability: Participants consistently reported high usability of the TrueAI. Notably, 88% of participants highlighted the streamlined workflow of the interface, with particular emphasis on the visually prominent pop-up. As participants (P2, P7, P9) illustrate this, "When I attempted to forward the news, the red pop-up immediately interrupted me and effectively captured my attention." This behavior intervention strategy, grounded in high-visibility design principles, effectively creates cognitive friction at key decision points, prompting older adults to pause and reassess the authenticity of the news.

Cognitive Changes: TrueAI's intervention effectively encouraged older adults to engage in proactive thinking before forwarding news. 75% of participants reported pausing to reflect and integrate their own judgment when encountering system prompts. This shift suggests that TrueAI not only achieves immediate behavioral interruption but also gradually cultivates critical thinking skills in users. As participant (P13) noted, "Previously, I didn't pay attention to the authenticity of news. From now on, I will be more vigilant about its credibility when reading."

5 Discussion and Limitation

This study introduces TrueAI, which provides prompts with low friction at high-significance key decision points to interrupt automatic forwarding behavior. It also offers functions for comparing true and fake news and for sharing authoritative news. These features collectively promote more reflective judgments.

While the findings are promising, this study has several limitations. The small and demographically narrow sample limits the broader applicability of the results. The evaluation took place in a simulated Chinese social media environment, which may not fully capture real-world complexities. Given the general inaccessibility of platform APIs, future work could explore client-side solutions, such as browser extensions, to enable local, privacy-preserving analysis. Additionally, the short-term intervention makes it difficult to assess long-term behavioral change. Future work should involve more diverse participants, real-world platforms, and extended study periods to better evaluate effectiveness, scalability, and sustained impact.

References

- [1] Iftikhar Ahmad, Muhammad Yousaf, Suhail Yousaf, and Muhammad Ovais Ahmad. 2020. Fake news detection using machine learning ensemble methods. *Complexity* 2020, 1 (2020), 8885861.
- [2] Hunt Allcott and Matthew Gentzkow. 2017. Social media and fake news in the 2016 election. *Journal of economic perspectives* 31, 2 (2017), 211–236.
- [3] Jennifer Allen, Cameron Martel, and David G Rand. 2022. Birds of a feather don't fact-check each other: Partisanship and the evaluation of news in Twitter's Birdwatch crowdsourced fact-checking program. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. 1–19.
- [4] Michelle A Amazeen and Bartosz W Wojdyski. 2019. Reducing native advertising deception: Revisiting the antecedents and consequences of persuasion knowledge in digital news contexts. *Mass Communication and Society* 22, 2 (2019), 222–247.
- [5] Michelle A Amazeen and Bartosz W Wojdyski. 2020. The effects of disclosure format on native advertising recognition and audience perceptions of legacy and online news publishers. *Journalism* 21, 12 (2020), 1965–1984.

- [6] Sumitra Badrinathan. 2021. Educative interventions to combat misinformation: Evidence from a field experiment in India. *American Political Science Review* 115, 4 (2021), 1325–1341.
- [7] Nadia M Brashier and Elizabeth J Marsh. 2020. Judging truth. *Annual review of psychology* 71, 1 (2020), 499–515.
- [8] Nadia M Brashier and Daniel L Schacter. 2020. Aging in an era of fake news. *Current directions in psychological science* 29, 3 (2020), 316–323.
- [9] Juan Cao, Peng Qi, Qiang Sheng, Tianyun Yang, Junbo Guo, and Jintao Li. 2020. Exploring the role of visual content in fake news detection. *Disinformation, Misinformation, and Fake News in Social Media: Emerging Research Challenges and Opportunities* (2020), 141–161.
- [10] John Cook, Stephan Lewandowsky, and Ullrich KH Ecker. 2017. Neutralizing misinformation through inoculation: Exposing misleading argumentation techniques reduces their influence. *PLoS one* 12, 5 (2017), e0175799.
- [11] Harris Ed Cooper, Paul M Camic, Debra L Long, AT Panter, David Ed Rindskopf, and Kenneth J Sher. 2012. *APA handbook of research methods in psychology; Vol 2: Research designs: Quantitative, qualitative, neuropsychological, and biological*. American Psychological Association.
- [12] Lisa Fazio. 2020. Pausing to consider why a headline is true or false can help reduce the sharing of false news. *Harvard Kennedy School Misinformation Review* 1, 2 (2020).
- [13] Nir Grinberg, Kenneth Joseph, Lisa Friedland, Briony Swire-Thompson, and David Lazer. 2019. Fake news on Twitter during the 2016 US presidential election. *Science* 363, 6425 (2019), 374–378.
- [14] Andrew Guess, Jonathan Nagler, and Joshua Tucker. 2019. Less than you think: Prevalence and predictors of fake news dissemination on Facebook. *Science advances* 5, 1 (2019), eaau4586.
- [15] Andrew M Guess, Brendan Nyhan, and Jason Reifler. 2020. Exposure to untrustworthy websites in the 2016 US election. *Nature human behaviour* 4, 5 (2020), 472–480.
- [16] Zhijiang Guo, Michael Schlichtkrull, and Andreas Vlachos. 2022. A Survey on Automated Fact-Checking. *Transactions of the Association for Computational Linguistics* 10 (2022), 178–206. doi:10.1162/tacl_a_00454
- [17] Anastasia Kozyreva, Stephan Lewandowsky, and Ralph Hertwig. 2020. Citizens versus the internet: Confronting digital challenges with cognitive tools. *Psychological Science in the Public Interest* 21, 3 (2020), 103–156.
- [18] Stephan Lewandowsky, John Cook, and Ullrich KH Ecker. 2017. Letting the gorilla emerge from the mist: Getting past post-truth. *Journal of Applied Research in Memory and Cognition* 6, 4 (2017), 418–424.
- [19] Stephan Lewandowsky, Ullrich KH Ecker, Colleen M Seifert, Norbert Schwarz, and John Cook. 2012. Misinformation and its correction: Continued influence and successful debiasing. *Psychological science in the public interest* 13, 3 (2012), 106–131.
- [20] James R Lewis. 2018. The system usability scale: past, present, and future. *International Journal of Human-Computer Interaction* 34, 7 (2018), 577–590.
- [21] Paul Mena. 2020. Cleaning up social media: The effect of warning labels on likelihood of sharing false news on Facebook. *Policy & internet* 12, 2 (2020), 165–183.
- [22] Ryan C Moore and Jeffrey T Hancock. 2022. A digital media literacy intervention for older adults improves resilience to fake news. *Scientific reports* 12, 1 (2022), 6008.
- [23] Tamara A Rahhal, Cynthia P May, and Lynn Hasher. 2002. Truth and character: Sources that older adults can remember. *Psychological Science* 13, 2 (2002), 101–105.
- [24] Dietram A Scheufele, Andrew J Hoffman, Liz Neeley, and Czerne M Reid. 2021. Misinformation about science in the public sphere. *Proceedings of the National Academy of Sciences* 118, 15 (2021), e2104068118.
- [25] Kellen Sharp, Rachel R Ouellette, Rujula Singh Rajendra Singh, Elise E DeVito, Neil Kamdar, Amanda de la Noval, Dhiraj Murthy, and Grace Kong. 2025. Generative artificial intelligence and machine learning methods to screen social media content. *PeerJ Computer Science* 11 (2025), e2710.
- [26] Tamara Sims, Andrew E Reed, and Dawn C Carr. 2017. Information and communication technology use is related to higher well-being among the oldest-old. *Journals of Gerontology Series B: Psychological Sciences and Social Sciences* 72, 5 (2017), 761–770.
- [27] Ian Skurnik, Carolyn Yoon, Denise C Park, and Norbert Schwarz. 2005. How warnings about false claims become recommendations. *Journal of Consumer Research* 31, 4 (2005), 713–724.
- [28] Soroush Vosoughi, Deb Roy, and Sinan Aral. 2018. The spread of true and false news online. *science* 359, 6380 (2018), 1146–1151.
- [29] Sam Wineburg and Sarah McGrew. 2019. Lateral reading and the nature of expertise: Reading less and learning more when evaluating digital information. *Teachers College Record* 121, 11 (2019), 1–40.
- [30] Dalal Youssef, Linda Abou-Abass, and Hamad Hassan. 2023. Unveiling the unknown: first comprehensive assessment of the knowledge, attitudes and practices of hospital cleaning services staff regarding COVID-19 in Lebanon during the pandemic. *Archives of Public Health* 81, 1 (2023), 134.